Miranda Muqing Miao

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Research Interests: Theory-grounded empirical analysis of how training data and training stages shape learned representations and behavior in deep networks; representation geometry; LLM alignment

EDUCATION

UNIVERSITY OF PENNSYLVANIA — **PHILADELPHIA, PA**

PhD in Computer Information Science (co-advised by Professor Lyle Ungar and Professor Michael Kearns) — Aug 2026 – Present

Master's in Computer Information Science (Continued into PhD) — Aug 2024 – Present

- GPA: 3.97/4.00
- Outstanding reviewer for the Mechanistic Interpretability Workshop at ICML 2026

WELLESLEY COLLEGE — **WELLESLEY, MA**

Bachelor of Arts in Data Science and Economics (Magna Cum Laude and Dept. Honors in Economics)

- GPA: 3.87/4.00; Durant Scholar | Study abroad: University of Cambridge (Math)

PUBLICATIONS & PREPRINTS

Hallucination, Monofacts, and Miscalibration: An Empirical Investigation

***Miranda Muqing Miao**, Michael Kearns* (Proceedings of the National Academy of Sciences: **PNAS 2025**)

A **theory-to-empirical** study providing the first empirical validation of the theoretical lower bound linking LLM hallucination rates to the monofact rate of the training data. Through controlled training runs that systematically vary the data distribution, shows that hallucination can be controlled by shaping the training distribution, and derives a selective upweighting method for mitigation from the theory

Conceptors for Semantic Steering

Ilias Triantafyllopoulos, **Miranda Muqing Miao**, et al. (**TACL 2026**)

A **mechanistic interpretability** paper recasting activation steering as a safety intervention. Models concepts as multidimensional subspaces of the residual stream via conceptor matrices. The closed-form Boolean algebra (AND, OR, NOT) on these subspaces lets AND-NOT suppress a harmful concept without interfering with unrelated capabilities, cutting degenerate outputs from 58% to 13%

The Impact of Language Mixing on Bilingual LLM Reasoning

*Yihao Li, Jiayi Xin, **Miranda Muqing Miao**, et al.* (**EMNLP 2025 Main**)

Traces language mixing in bilingual reasoning models to a specific post-training stage (RLVR), isolating how a single training stage reshapes learned behavior. Shows that mixing enhances mathematical accuracy and develops a lightweight probe to guide beneficial code-switching during decoding

Correctness-Optimized Residual Activation Lens (CORAL): Transferrable and Calibration-Aware Inference-Time Steering

***Miranda Muqing Miao**, Young-Min Cho, Lyle Ungar* (**ICML 2026**)

A **mechanistic interpretability** approach to inference-time steering that extracts distributed correctness signals from frozen LLM residual stream activations via auxiliary probes, improving accuracy and calibration, with the learned correctness directions transferring across tasks without any weight updates

Closing the Confidence-Faithfulness Gap in Large Language Models

***Miranda Muqing Miao**, Lyle Ungar*

A **mechanistic interpretability** paper on verbalized confidence faithfulness, showing that calibration and verbalized confidence are linearly encoded in orthogonal subspaces across three models. Identified the "Reasoning Contamination Effect" and introduced a two-stage adaptive steering pipeline that substantially improves calibration alignment

COAST: Unlocking Vision-Language-Action Models through Hidden States

Miranda Muqing Miao*, *Subin Kim*, Brandon Yang, Lyle Ungar* (**ICML 2026 Mech Interp. Workshop)**

A **mechanistic interpretability** extension to robotics, applying contrastive activation steering to frozen VLA policy internals, recovering manipulation performance at inference time without parameter updates while understanding VLA representations

Multilingual Retrieval Augmented Generation for Culturally-Sensitive Tasks: A Benchmark for Cross-lingual Robustness

*Bryan Li, Fiona Luo, Samar Haider, Adwait Agashe, Siyu Li, Runqi Liu, **Miranda Muqing Miao**, Shriya Ramakrishnan, Yuan Yuan, Chris Callison-Burch* (**ACL 2025 Findings**)

Contributes to BordIRLines, a multilingual RAG benchmark for territorial disputes across 49 languages, demonstrating that cross-lingual retrieval improves response consistency and reduces geopolitical bias compared to monolingual approaches

ONGOING PROJECTS

SPAR Research Fellowship Fall 2026

Mentored by Noah Siegel (Google DeepMind, UCL) on training LLMs to improve the faithfulness of self-explanations, as measured by Chain-of-Thought simulatability. Selected as a member of the 10 people cohort out of 500, at an acceptance rate of 2%

Working PhD Thesis: "From Individual to Collective Alignment in Multi-Agent LLM Systems"

My work studies when a multi-agent LLM system's behavior can be predicted and kept aligned from the properties of its individual agents.

RELEVANT PROFESSIONAL EXPERIENCES

UBER — **NEW YORK CITY, NY**
AI Engineer on core ML team — JUNE - AUGUST 2026

EDGEPEAK CAPITAL (STEALTH STARTUP) — **NEW YORK, NY**
Agentic AI Researcher — MAY 2025 – MAY 2026